

ibm360_disk_sop_v1json

```
{
  "id": "SOP-IBM360-003",
  "title": "IBM 2311/2314 Disk Storage Unit 'Unit Check' Diagnostic",
  "system": "IBM System/360",
  "component": "2311/2314 Direct Access Storage Device (DASD)",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY22-2841 / Y26-5897)",
  "description": "Diagnostic procedure for isolating hardware errors reported as 'Unit Check' in the Channel Status Word (CSW).",
  "steps": [
    {
      "step": 1,
      "action": "CSW Error Validation",
      "details": "When a 'Unit Check' is indicated in the CSW, examine the Status Portion (bits 32-47). Check for bit 38 (Unit Check) and bit 39 (Unit Exception).",
      "check": "Unit Check bit must be 1 to proceed with Sense data analysis."
    },
    {
      "step": 2,
      "action": "Sense Command Execution",
      "details": "Issue a Sense command (Command Code 04) to the failing device address to retrieve the 6-byte Sense Data.",
      "logic": "The first 24 bits (Bytes 0, 1, 2) contain the critical diagnostic indicators."
    },
    {
      "step": 3,
      "action": "Byte 0 Analysis (Command/Data Errors)",
      "details": "Examine Byte 0 of the Sense Data.",
      "indicators": {
        "Bit 0 (Command Reject)": "The control unit cannot execute the command. Check for invalid seek address or write-protected disk pack.",
        "Bit 1 (Intervention Required)": "The drive is not ready. Check if the disk pack is spinning or if the 'Start/Stop' switch is in the correct position.",
        "Bit 2 (Bus Out Parity Check)": "Electrical error on the interface cables. Check cable seating and terminations.",
        "Bit 3 (Equipment Check)": "Internal hardware failure in the drive or control unit. Proceed to internal diagnostic tests."
      }
    },
    {
      "step": 4,
      "action": "Byte 1 Analysis (Seek/Data Errors)",
      "details": "Examine Byte 1 for seek-specific issues.",
      "indicators": {
        "Bit 0 (Data Check)": "A parity or cyclic check error occurred during a read or write operation. Retry the operation.",
        "Bit 2 (Seek Check)": "The access mechanism failed to reach the correct cylinder. Recalibrate (Seek to cylinder 0) and retry."
      }
    }
  ]
}
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- ,
"diagnostic_flowchart": [
  {
    "condition": "Intervention Required bit ON",
    "next": "Check drive ready lights and pack seating."
  },
  {
    "condition": "Command Reject bit ON",
    "next": "Check pack write-protect switch and CCW (Channel Command Word) validity."
  },
  {
    "condition": "Data Check bit ON",
    "next": "Perform automated error recovery (retries). If it fails after 10 tries,
perform head cleaning or pack replacement."
  }
]
}

```